Project INSPIRE Design Document

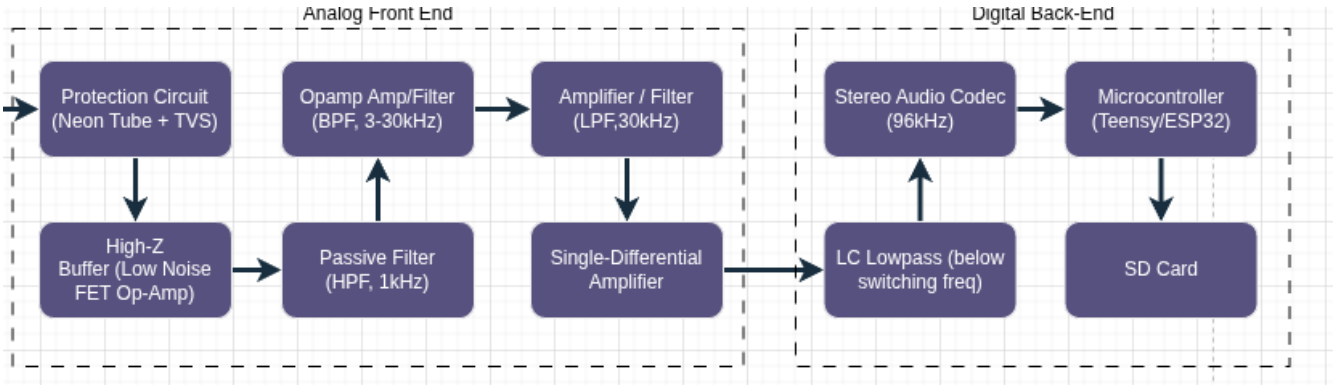
Project Part V: Active Low-Pass Filter

Revision 1.0: Presented 10/9/2026

1) General Project Specifications

The purpose of the project is to build a receiver to capture natural radio atmospheric phenomena in the Very Low Frequency (VLF) range of 3-30kHz.

A block diagram of the project is provided below:



The project will be built as two Printed Circuit Boards (PCBs) inside shielded metal enclosures, and powered off batteries, in order to establish isolation and distance from power lines, which are very large emitters of interference.

In order to pick up these signals, a high resolution 24-bit audio Analog-Digital Converter (ADC) is required. As such, we can establish a dynamic range of approximately 100dB. This means that the analog stage should be prioritized to ensure linearity and prevent clipping.

2) Sub-circuit Specifications

The analog Low-Pass Filter (LPF) serves a special role, which is to act as a final amplifier stage, and also to cutoff all frequencies above 30kHz. If any signals leak through, even though they are past the Nyquist frequency of 48kHz, they can alias and appear in our measurements. As such, a steep LPF is required. The following parameters need to be met:

- This stage will implement a higher-order active low-pass filter (such as a second-order LPF) to achieve a maximally flat passband up to 30kHz and a steep, rapid roll-off beyond the cutoff.
- Gain Stage Tuning: This block provides the second 10dB of gain.
- Voltage Restrictions: This system needs to work with the 3.3V dual-rail power supply.

3) Expected Outputs

This sub-circuit design requires a technical write-up and simulation covering three major parts:

- **Topology & Component Values:** A justification of filter topology (with appropriate schematic) and component values must be determined first. This includes existing cutoff values of 30kHz and the gain of 10dB.
- **Op-Amp Selection:** Find a suitable operational amplifier, and give a justification for it, along with relevant parameters, like GBW, Slew Rate, and output impedance.
- **SPICE Simulation:** A two-stage SPICE simulation is required:
    - An AC analysis showing the filter response well past the region of interest. The rolloff needs to be as expected past 30kHz, and the gain of +10dB needs to be present in the pass-band.
    - A Noise analysis needs to be run to ensure this element is not providing excess noise to the system.

Optional Extra (Stretch Goals):

- Try a higher-order filter, like a 4th order Chebyshev or Butterworth filter! These require dual-opamp circuits, but double the effective roll-off to 24dB/octave, or 40dB/decade.